

We need to specify this base in our directory configuration file along with the superuser name and password.

The superuser is referred to as *rootdn* (the DN of the 'root' user). This should also fall under the base of the directory tree and could be referred to by the Common Name (cn) of *Manager*. Finally, we should set the password of the root user.

We can also set the rootdn password in clear text. This is a security risk and should be avoided. The password should be stored in an encrypted form. I am only including the 'text password' for demo purposes.

```
#####
#Sample /etc/openldap/slapd.conf
database bdb
suffix "dc=knafl,dc=org"
rootdn "cn=Manager,dc=knafl,dc=org"
# Cleartext passwords, especially for the rootdn,
# should be avoided.
# See slapdpasswd(8) and slapd.conf(5) for details.
# Use of strong authentication encouraged.
rootpw FOSTERngLauX
#####
```

Apart from the above, I have made no changes to the default configuration file. Once we have set up the server, we need to start the LDAP daemon. The LDAP service will read the configuration file and start the *slapd* daemon.

```
[root@vbg openldap]# service ldap start
Checking configuration files for slapd: bdb db open: Warning - No
DB_CONFIG file found in directory /var/lib/ldap: (2)
Expect poor performance for suffix dc=knafl,dc=org.
config file testing succeeded

Starting slapd: [ OK ]
[root@vbg openldap]#
```

To check whether our LDAP is up, we will query our server as follows:

```
[root@vbg openldap]# ldapsearch -x -h localhost -b "dc=knafl,dc=org"
# extended LDIF
#
# LDAPv3
# base <dc=knafl,dc=org> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# search result
# search: 2
# result: 32 No such object
#
# numResponses: 1
[root@vbg openldap]#
```

We will receive the 'No such object' message from the server. This means that our LDAP server is running, but it could not locate our directory tree or the data.

Now we need to create the base data for our LDAP server. To do that, we need to create an LDIF file and use the client command *ldapadd*.

First, let us create a simple LDIF file for our directory base and save it in the file */tmp/ldap-base*:

```
[root@vbg ~]# cat /tmp/ldap-base
dn: dc=knafl,dc=org
dc: knafl
objectClass: top
objectClass: domain
```

Adding the entries to the directory:

```
[root@vbg ~]# ldapadd -x -D "cn=Manager,dc=knafl,dc=org" \
-W -f /tmp/ldap-base
Enter LDAP Password:
adding new entry 'dc=knafl,dc=org'

[root@vbg ~]#
```

By default, we require rootdn privileges to write data in the directory. The *-D* option of the *ldapadd* command specifies the username (DN in LDAP terminology), and *-W* specifies that the password will be entered on the command line prompt. Finally, the *-f* option specifies the name of the LDIF file that contains entries to be added to the directory.

Now when we query the directory, we get an output as follows:

```
root@vbg ~]# ldapsearch -x -h localhost -b "dc=knafl,dc=org"
# extended LDIF
#
# LDAPv3
# base <dc=knafl,dc=org> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# knafl.org
dn: dc=knafl,dc=org
dc: knafl
objectClass: top
objectClass: domain
#
# search result
search: 2
result: 0 Success
# numResponses: 2
# numEntries: 1
```

Thus we have created the base of our directory. Now we need to add more data to it and also beef up the security of the LDAP server.